

A Book on

*The LLM Odyssey: Foundations, Architectures, and
Beyond Building with Generative AI: A Hands-on
Textbook for LLM Practitioners*

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Declaration of Authorship

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other University. This dissertation is the result of my own work and includes nothing which is the outcome of work done in collaboration, except where specifically indicated in the text.

Signed:

Date:

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Abstract

Contents

Declaration of Authorship	i
Acknowledgements	ii
Abstract	iii
Contents	iv
List of Figures	vi
List of Tables	vii
Abbreviations	viii
Definitions	ix
1 Introduction	1
1.1 Introduction	1
1.2 Research issues	1
1.3 Objective	1
1.4 Contributions	1
1.5 Papers published/communicated based on this research	1
1.6 Organization of thesis	1
2 Literature Review	2
3	3
3.1 Introduction	3
4	4
5	5

6 Conclusion and Future Direction	6
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Bibliography	7
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List of Figures

List of Tables

Abbreviations

CNN	C onvolut i onal N eural N etwork
GB	G radient B oosting
LSTM	L ong S hort T erm M emory
NLTK	N atural L anguage T oolkit
POS	P arts o f S peech
RF	R andom F orest
RNN	R ecurrent N eural N etwork
ROC	R eciever O perating C haracteristics
SVM	S upport V ector M achine

Definitions

Classification	Separating data into groups of classes.
Precision	Precision is the ratio of a number of events we can correctly identify to a number of all events we identify.
Recall	Recall is the ratio of a number of events we can correctly classify to a number of all correct events.
F₁-score	F ₁ -score is the harmonic mean of Precision and Recall.

Dedicated to my parents. . .

Chapter 1

Introduction

The objective of this chapter is:

- 1.

1.1 Introduction

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1.2 Research issues

1.3 Objective

1.4 Contributions

1.5 Papers published/communicated based on this research

1.6 Organization of thesis

Chapter 2

Literature Review

Chapter 3

The objectives of this chapter are following:

3.1 Introduction

Chapter 4

The objectives of this chapter are following:

Chapter 5

The objectives of this chapter are following:

- 1.

Chapter 6

Conclusion and Future Direction

The objectives of the chapter are:

1. To summarize the work done.
2. To suggest future directions.

Bibliography

Publications from this thesis

- Journals:
- Conference proceedings: